

LUCIDPREP

Grade 12 Physics

U.S. Standards-Aligned Course Syllabus

PHYSICS

Physics · NGSS

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

Physics is the most common Grade 12 science course in the standard Biology → Chemistry → Physics sequence, though a fourth year of science is not required in every state. This Physics course mirrors the Physics option in the Grade 9 curriculum; a school places it once, in whichever grade its sequence uses. The NGSS are the dominant national reference.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

This 36-week course is designed around **the NGSS High School Physical Science standards for physics (HS-PS2 motion & stability, HS-PS3 energy, HS-PS4 waves), with Science & Engineering Practices and Crosscutting Concepts**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 12
Course	Physics · NGSS
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Model motion, forces, energy and momentum mathematically and conceptually.
- Apply Newton's laws and conservation principles.
- Analyze waves, sound, light and information transfer.
- Model electrical and magnetic systems.
- Use equations, graphs, experiments and evidence to explain physical phenomena.

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Module 1 — Motion, Measurement & Forces

Week	Topics	Students will be able to	Standards
Week 1 — Physics & Measurement	SI units; measurement; uncertainty; scalars/vectors; graphs	Use units; distinguish scalar/vector; interpret graphs	SEP/modeling
Week 2 — Describing Motion	Position; distance; displacement; time; speed; velocity	Calculate and interpret motion	HS-PS2
Week 3 — Acceleration	Acceleration; velocity-time and position-time graphs; free fall	Calculate acceleration; interpret graphs	HS-PS2
Week 4 — Forces & Free-Body Diagrams	Net force; gravity; normal; friction; contact/non-contact forces	Draw FBDs; calculate net force	HS-PS2-1/2
Week 5 — Newton's Laws	First/second/third laws; inertia; mass; force	Apply Newton's laws	HS-PS2
Week 6 — Friction & Applications	Static/kinetic friction; drag; engineering applications	Analyze friction and design tradeoffs	HS-PS2
Week 7 — Gravity & Circular Motion	Universal gravitation; mass vs weight; orbits; centripetal acceleration	Model gravitational/circular systems	HS-PS2-4
Week 8 — Motion Investigation	Motion sensors; data tables; graph fitting; evidence	Design and analyze investigation	SEP
Week 9 — Module Assessment	Motion; acceleration; forces; Newton; gravity	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

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Module 2 — Energy, Work & Momentum

Week	Topics	Students will be able to	Standards
Week 1 — Work & Energy	Work; energy; force/displacement; kinetic/potential energy	Calculate work; connect work and energy	HS-PS3
Week 2 — Kinetic & Potential Energy	Kinetic; gravitational; elastic; energy diagrams	Calculate/model energy	HS-PS3-1
Week 3 — Conservation of Energy	Energy transfer; mechanical energy; thermal energy; systems	Track energy and explain conservation	HS-PS3
Week 4 — Power & Efficiency	Power; energy/time; efficiency; machines	Calculate power; evaluate efficiency	HS-PS3
Week 5 — Momentum	Momentum; impulse; force-time	Calculate momentum; explain impulse	HS-PS2-2
Week 6 — Collisions	Elastic/inelastic; momentum conservation; systems	Analyze collisions	HS-PS2-2
Week 7 — Energy & Momentum Engineering	Vehicle safety; crumple zones; airbags; sports; rockets	Apply conservation laws to design	Engineering/PS2/PS3
Week 8 — Investigation	Experimental design; data; uncertainty; CER	Conduct and explain investigation	SEP/CCC
Week 9 — Module Assessment	Work; energy; power; momentum; collisions	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

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Module 3 — Waves, Sound, Light & Information

Week	Topics	Students will be able to	Standards
Week 1 — Wave Properties	Mechanical/EM waves; amplitude; wavelength; frequency; period; speed	Describe and calculate wave properties	HS-PS4
Week 2 — Sound Waves	Longitudinal waves; pitch; loudness; speed; resonance	Connect frequency to pitch; analyze sound	HS-PS4
Week 3 — Light & EM Waves	EM spectrum; frequency; wavelength; energy; visible light	Compare EM waves and applications	HS-PS4-1
Week 4 — Reflection & Refraction	Reflection; refraction; mirrors; lenses; ray models	Predict optical behavior	HS-PS4
Week 5 — Wave Interactions	Interference; diffraction; superposition; standing waves; resonance	Model wave interactions	HS-PS4
Week 6 — Communication & Digital Information	Signals; analog/digital; encoding; bandwidth; noise	Explain wave-based information transfer	HS-PS4
Week 7 — Waves in Technology	Fiber optics; radio; Wi-Fi; imaging; remote sensing	Connect wave behavior to technology	HS-PS4 applications
Week 8 — Wave Investigation	Measurements; graphs; model fitting; evidence	Measure and model wave behavior	SEP
Week 9 — Module Assessment	Waves; sound; light; optics; information	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

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Module 4 — Electricity, Magnetism & Modern Physics

Week	Topics	Students will be able to	Standards
Week 1 — Electric Charge & Fields	Charge; electric force; fields; conductors; insulators	Model electric interactions	HS-PS2
Week 2 — Electric Potential & Circuits	Voltage; current; resistance; series/parallel circuits	Interpret and calculate basic circuits	PS3/PS2
Week 3 — Ohm's Law & Electrical Power	Resistance; Ohm's law; power; household energy	Solve circuit problems; calculate power	Quantitative physical science
Week 4 — Magnetism	Magnetic fields; current; electromagnets; magnetic force	Model magnetic fields and electromagnets	HS-PS2
Week 5 — Electromagnetic Induction	Induction; generators; motors; transformers	Explain energy conversion	PS2/PS3
Week 6 — Modern Physics Foundations	Photons; quantum idea; nuclear energy; relativity overview	Explain limits of classical models	High-school extension
Week 7 — Engineering Design Challenge	Constraints; energy; forces; waves/electricity; optimization	Test and improve a design	NGSS Engineering Design
Week 8 — Physics Review & Synthesis	Mechanics; energy; momentum; waves; electricity; magnetism	Connect principles and solve novel problems	Course synthesis
Week 9 — Final Assessment	Comprehensive quantitative/data/model assessment	Demonstrate course mastery	Final performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

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Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	NGSS: HS-PS2, HS-PS3, HS-PS4
Science & Engineering Practices	Asking questions and defining problems; developing and using models; planning and carrying out investigations; analyzing and interpreting data; using mathematics and computational thinking; constructing explanations and designing solutions; engaging in argument from evidence; obtaining, evaluating and communicating information.
Crosscutting Concepts	Patterns; cause and effect; scale, proportion and quantity; systems and system models; energy and matter; structure and function; stability and change.
State mapping	NGSS adopted or adapted by nearly all states; middle-school standards are banded 6–8 and arranged differently by state.

Content Production Guidance

Author each week around an anchoring phenomenon or problem: 5 warm-ups, an explanatory lecture with models, 2 videos, an AI presentation with visual models and investigations, 2 homework sets, and a 40-question bank mixing three-dimensional items (SEP × DCI × CCC). Tag items to NGSS performance expectations.

Curriculum Basis & Source Notes

This Grade 12 science syllabus was generated from the Next Generation Science Standards. It reflects Physics, the common Grade 12 science course; the same content appears as the Physics option in the Grade 9 curriculum, since HS science placement varies by district.

Primary reference	Next Generation Science Standards — HS Physical Science performance expectations for physics (HS-PS2, HS-PS3, HS-PS4), with Science & Engineering Practices and Crosscutting Concepts.
Framework basis	A Framework for K-12 Science Education (National Research Council).
State cross-check	California and other NGSS-adopting state science frameworks.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Select anchoring phenomena per module and commission three-dimensional item banks (SEP × DCI × CCC) tagged to NGSS performance expectations.